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AI in Military

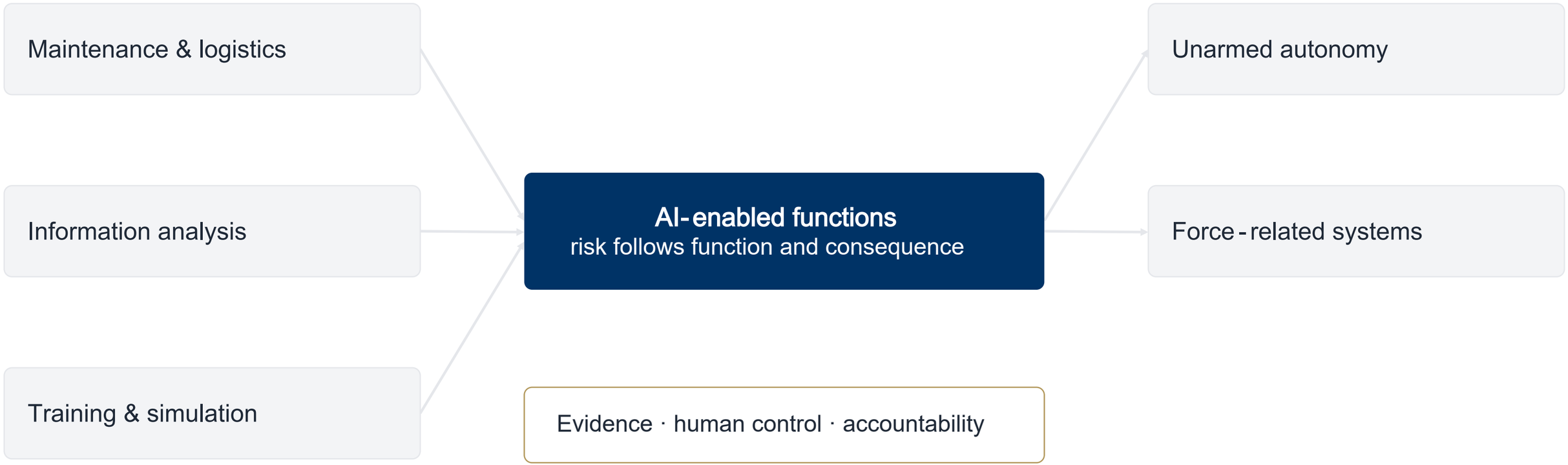
Sirui Han · HKUST

MAIE5103 Artificial Intelligence Ethics · Lecture 9 / Week 10

Systems · human judgment · assurance · procurement

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Military AI extends beyond autonomous weapons



Scope framing: ICRC position on autonomous weapon systems (12 May 2021); NATO AI Strategy (10 Jul 2024).

Sources: ICRC; NATO · Functional map is conceptual, not an operational workflow.

AI, autonomy, and unmanned platforms are different

Term	Core idea	Does not automatically mean	High-level example
AI	Computational prediction, classification, or generation.	Autonomy, physical embodiment, or independent force.	Maintenance forecast
Autonomy	A function is delegated under defined human arrangements.	Machine learning, or absence of human responsibility.	Navigation function
Unmanned	No person is physically onboard a platform.	Autonomy, weapon status, or lack of remote control.	Remotely operated vehicle

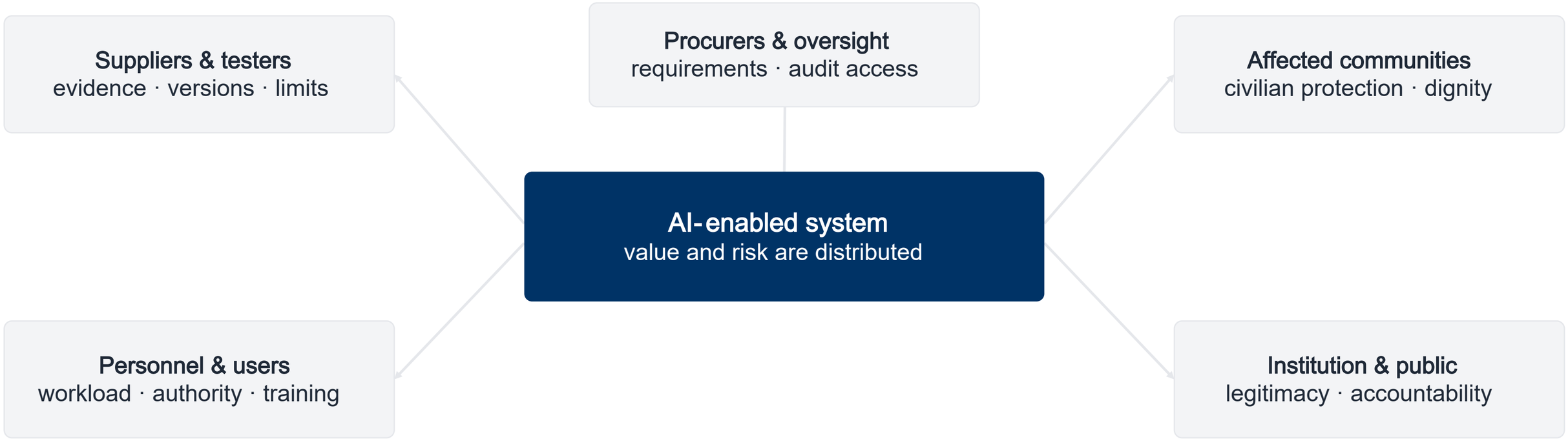
An unarmed autonomous vehicle is not automatically a weapon; a remotely controlled system is not automatically autonomous.

Source: ICRC, Position on Autonomous Weapon Systems, 12 May 2021.

Definitions distinguish a function, a system configuration, and an onboard-personnel distinction.

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The customer is not the only affected stakeholder



Ethics framing: ICRC (2021) and NATO Responsible Use principles (2024).

Accountability spans organizations; the map is illustrative, not a legal allocation of liability.

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Law governs conflict and conduct at different levels

Jus ad bellum

Question addressed: authority to resort to force.

This is distinct from the rules governing conduct during an armed conflict.

Jus in bello / IHL

Question addressed: conduct during armed conflict.

Civilian-protection principles: distinction, proportionality, and feasible precautions.

High-level orientation only: not an incident classification, targeting procedure, or numerical harm calculation.

Sources: ICRC, What is international humanitarian law? (2022); Customary IHL Rules 1, 14, 15.

IHL frames conduct in conflict; it does not itself answer whether force may be resorted to.

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A prediction score is not a legal judgment

A statistical signal can inform review; it cannot decide legality or ethical legitimacy.

Prediction score

- Pattern-based estimate
- Depends on data, model, and stated scope
- May be uncertain or wrong

Legal judgment

- Context and applicable rules
- Human responsibility and reasons
- Not reducible to accuracy

Ethical legitimacy

- Accountability and dignity
- Civilian protection and trust
- Institutional justification

Sources: ICRC Customary IHL Rules 1, 14, 15; ICRC, What is IHL? (2022).

No numerical harm calculus is implied; legal judgment remains context-specific.

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Project Maven began as analyst support (2017)

26 APRIL 2017

Deputy Secretary memorandum establishes the Algorithmic Warfare Cross-Functional Team / Project Maven.

Primary memorandum (PDF)

Initial purpose
video-analysis support to reduce analyst workload

Human & institutional responsibility
review, authority, and accountable decisions remain distinct

Analyst-workload support does not establish autonomous lethal authority; it also does not remove downstream ethical stakes.

Source: U.S. DoD, Deputy Secretary memorandum, 26 Apr 2017. Native source-anchor diagram; no memo image is used.

Claim scope: establishment and stated initial support purpose, not proof of later program continuity or authority.

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Employee voice can change a supplier decision

2017
Project Maven begins with an analyst-support purpose.

→

Workforce voice
Employee opposition becomes a governance input for the supplier.

Reuters reports the decision; no employee counts are needed here.

→

1 JUN 2018
Reuters reports Google would not renew the contract while honoring its remainder.

Supplier exit is not the same as program end; workforce voice can reshape a business decision.

Source: Reuters, 1 Jun 2018, “Google to scrub U.S. military deal protested by employees,” reporting a non-renewal decision.

Reporting establishes the reported supplier decision, not an immediate cancellation or the end of Project Maven.

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A contract award is not evidence of public value

29 MAY 2024
Official notice: Palantir USG Maven Smart System prototype.

$480 million
announced contract value

Funding by order
Work locations and funding determined with each order.

Estimated completion
28 May 2029

The notice does not report expenditure, validate performance, or explicitly describe a formal ceiling.

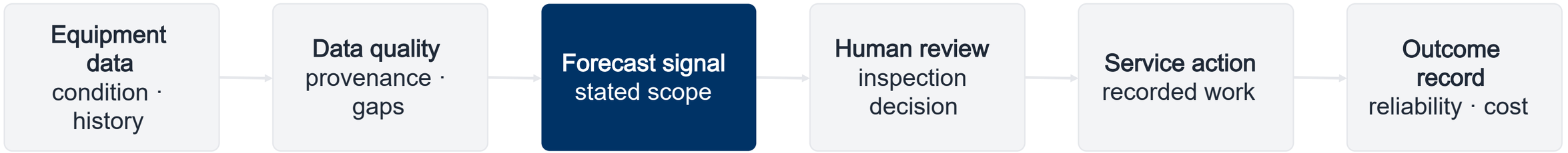
Source: U.S. Department of Defense contract announcement, 29 May 2024.

Scope: an announced award is procurement evidence, not a performance audit or public-value finding.

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Predictive maintenance offers a bounded business case

Illustrative noncombat maintenance analytics: reliability, availability, and cost must be evaluated together.



Evidence to retain: data provenance · model/version records · inspection decision · outcome and cost measure.

Context: GAO-23-105556, 8 Dec 2022. Flow is conceptual, not a description of a Navy pilot.

Analytics can support maintenance planning; a forecast does not replace inspection or evaluation.

GAO: pilot targets were not demonstrated benefits

Navy pilot goals reported by GAO; percentages below are targets, not achieved results.

Navy pilot	Reactive maintenance target	Targeted increase in predictive maintenance	GAO finding, 2022
F/A-18 Super Hornet	20%	26%	Repeatable evaluation plan lacking
DDG-51	38%	24%	Repeatable evaluation plan lacking

GAO found no defined, repeatable evaluation plan. Pilot goals are not measured savings, baselines, or demonstrated outcomes.

Source: U.S. GAO-23-105556, 8 Dec 2022, Figure 7 (report p. 31).

GAO assessed pilots and evaluation practices; the report does not substantiate vendor ROI or a complete project inventory.

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Maintenance value requires a credible comparison

A value claim is an evaluation design, not a return-on-investment headline.

1 · Baseline

Comparable work, equipment state, and observation period are defined before comparison.

2 · Outcomes

Reliability, maintenance burden, alerts, and realized work are recorded consistently.

3 · Full burden

Labor, inspection, parts, rework, data quality, and lifecycle costs remain visible.

4 · Bounded decision

Assess net value and limits against the baseline. Preserve uncertainty and safety constraints.

Case evidence, 8 Dec 2022: GAO found inconsistent tracking and insufficient repeatable evaluation in military predictive-maintenance pilots. It did not report a validated ROI.

Source: U.S. GAO, GAO-23-105556, 8 Dec 2022 · Proposed evaluation design; maintenance scope only.

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Assurance evaluates a system, not just a model

EDITABLE ASSURANCE LAYERS

Organization: ownership, review, and escalation

Interface and workflow: status, roles, and handoffs

Model and system: stated requirements and behavior

Data: provenance, coverage, quality, and change history

Evidence must match the claim

V&V: evidence against stated hardware and software requirements.

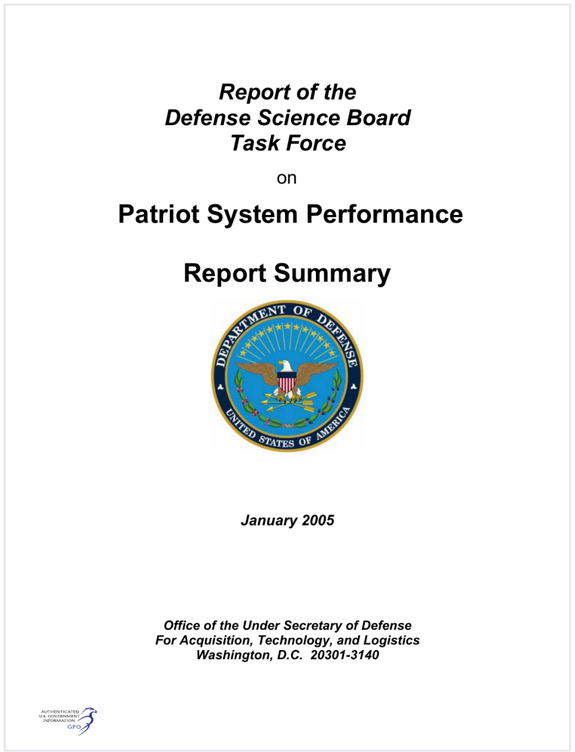
T&E: evidence under realistic, documented conditions —not only a benchmark.

Governance: evidence of who can interpret, intervene, and review change.

Generic assurance framing: a passing model metric cannot, by itself, substantiate reliability, human control, or organization al readiness.

Sources: U.S. DoDD 3000.09, 25 Jan 2023 · NATO revised AI strategy, 10 Jul 2024.

Patriot (2003): a conventional-automation analogue



Report of the
Defense Science Board
Task Force

on

Patriot System Performance

Report Summary

January 2005

Office of the Under Secretary of Defense
For Acquisition, Technology, and Logistics
Washington, D.C. 20301-3140

Public DSB summary cover, Jan 2005.
Provenance, not performance data.

23 Mar 2003 (local) — Tornado incident. A later UK review described multiple contributory factors.

2 Apr 2003 — F/A-18 incident, contemporaneously reported and included in the DSB's two-incident account.

Jan 2005 — DSB public summary: two Patriot firings at coalition aircraft; three aircrew deaths.

Bounded lesson: this was conventional automation, not documented modern ML. The DSB emphasized multifactored control, trust, interface, training, and information-sharing issues —not operator blame.

Sources: Defense Science Board public summary, Jan 2005 · UK MoD accident summary, May 2004 · CBS, 2 Apr 2003.

Formal oversight can lack practical control



Human override ≠ a safety guarantee; presence alone does not establish meaningful control.

Sources: DSB public summary, Jan 2005 · Hawley, Human Factors, 2008 · Conceptual control-capacity framing.

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DARPA ACE: moving from simulation to live flight



X-62A VISTA context photo; published Feb 2023. Credit: Kyle Brasier / U.S. Air Force.

2020 · AlphaDogfight was a virtual simulation event, not a live-aircraft outcome.

Dec 2022 · First ACE software live flights; an onboard safety pilot was retained.

Sept 2023 · Controlled X-62A test milestone, announced Apr 2024; safety pilots retained independent disengage authority.

Research and flight demonstration—not operational deployment, a safety guarantee, or lethal-decision authority.

Full source video · DARPA / U.S. Air Force test perspective

Sources: DARPA, 2020; 13 Feb 2023; 17 Apr 2024 · Aerospace Testing International, 14 Feb 2023.

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The sim-to-real boundary changes the safety claim

EVIDENCE CAN INCREASE IN REALISM WHILE REMAINING BOUNDED

Simulation

Evidence depends on modeled assumptions and scenarios.

Testbed

Real system elements introduce physical variation and safeguards.

Conditions remain selected.

Controlled flight

Evidence is closer to physical conditions, with documented constraints and safety personnel.

It is still a bounded demonstration.

Claim boundary

Does not establish:

- general safety
- operational deployment suitability
- authority to apply force

More realism narrows some uncertainty; it does not erase the need to state what was tested.

Sources: DARPA ACE, 2020; 13 Feb 2023; 17 Apr 2024 · Conceptual evidence ladder, not an acceptance standard.

Benchmark performance has a defined scope

Metrics are evidence for a stated condition. They are not portable claims of safety or authority.

Metric or record	What it can support	What it does not establish
Simulation outcome	Behavior within modeled conditions	Physical generalization beyond those conditions
Test completion	Completion of a documented test scope	Open-ended safety in other contexts
Human-intervention record	A recorded demonstration condition	A safety guarantee or meaningful control everywhere
Generalization evidence	Support only for documented variation	Universal performance or lethal -authority validation

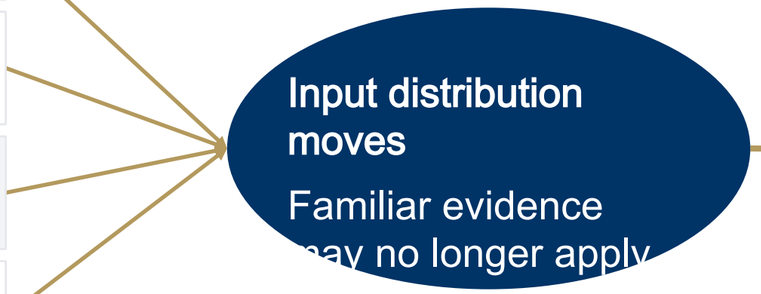
Interpretive bounds, not ACE results: a table of evidence categories rather than performance values.

Sources: DARPA ACE public descriptions, 2020, 13 Feb 2023, and 17 Apr 2024 · Scope framing by author.

Distribution shift can invalidate familiar evidence

GENERIC STRESSORS

- Environment and physical conditions
- Data quality and missingness
- Data integrity disruption
- Context or usage change



Assurance response

Detect validity signals.

Record condition and version.

Bound or pause use while evidence is investigated.

Re-evaluate against changed conditions.

Distribution shift is an engineering assurance risk, not an attack scenario.

Source: Conceptual engineering risk model · DoDD 3000.09, 25 Jan 2023, calls for realistic test and evaluation.

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Uncertainty must remain visible to decision-makers

Conceptual evidence display	
Evidence signal	Observed record and its provenance
Confidence	Range and calibration status, not a command
Known limits	Coverage gaps and change since validation
Review boundary	Abstain or seek independent review when evidence is inadequate

Visible uncertainty enables

Traceable reasons for a view.

Explicit limits and missing evidence.

A record of what could change the view.

Independent review before a consequential decision.

Confidence ≠ correctness.

Conceptual display only—not a deployment interface or a substitute for accountable judgment.

Source: Conceptual assurance display · DoDD 3000.09, 25 Jan 2023, addresses clear status, roles, and auditable methods.

Efficiency benefits do not settle ethical trade- offs

Before scale, each value claim needs countervailing risks and evidence for responsible use.

Benefit sought	Ethical tension	Evidence before scaling
Workload relief	Automation bias or a hidden transfer of burden to exceptions	Recorded exceptions and independent user feedback
Reliability	Missed or false alerts can redistribute risk and accountability	Comparison against baseline, lifecycle costs, and change controls
Consistency	Edge cases and affected people can be obscured, including in high -consequence settings	Stress evidence, accountable review, and transparency appropriate to scope

Proposed trade -off matrix: efficiency does not itself demonstrate human control, civilian protection, or institutional legitimac y.

Sources: ICRC position on autonomous weapon systems, May 2021 · DoDD 3000.09, 25 Jan 2023.

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Intermission

From support and assurance to autonomy, claims, and governance

Evidence boundaries remain central as the lecture turns to consequential delegation and accountability.

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Autonomy delegates a consequential function

Human authority
Purpose and constraints remain institutionally owned.

→

Delegated function
Autonomy concerns the function, not the platform label.

→

Accountable governance
Scope, intervention and review must remain meaningful.

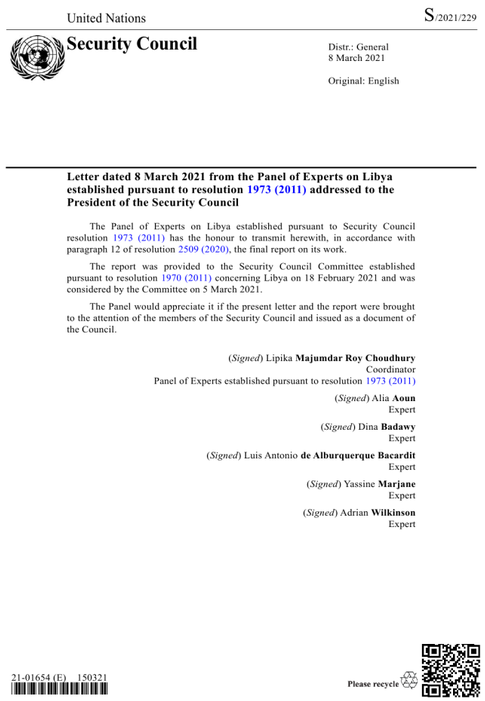
Control requirements: explicit scope · intervention authority · reviewable records

ICRC lens: autonomous weapon systems select and apply force without human intervention.

Source: ICRC, Position on Autonomous Weapon Systems, 12 May 2021. Conceptual allocation of authority; not an operating sequence.

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Libya (2021): the primary report has limits



United Nations S/2021/229

Security Council

Distr.: General
8 March 2021

Original: English

Letter dated 8 March 2021 from the Panel of Experts on Libya established pursuant to resolution 1973 (2011) addressed to the President of the Security Council

The Panel of Experts on Libya established pursuant to Security Council resolution 1973 (2011) has the honour to transmit herewith, in accordance with paragraph 12 of resolution 2509 (2020), the final report on its work.

The report was provided to the Security Council Committee established pursuant to resolution 1970 (2011) concerning Libya on 18 February 2021 and was considered by the Committee on 5 March 2021.

The Panel would appreciate it if the present letter and the report were brought to the attention of the members of the Security Council and issued as a document of the Council.

(*Signed*) Lipika **Majumdar Roy Choudhury**
Coordinator
Panel of Experts established pursuant to resolution 1973 (2011)

(*Signed*) Alia **Aoun**
Expert

(*Signed*) Dina **Badawy**
Expert

(*Signed*) Luis Antonio **de Alburquerque Bacardit**
Expert

(*Signed*) Yassine **Marjane**
Expert

(*Signed*) Adrian **Wilkinson**
Expert

21-01654 (E) 150321

Please recycle

UN Panel of Experts report · 8 Mar 2021

Report provenance and qualified mention

Paragraph 63 (PDF p. 17) uses mixed wording that mentions Kargu -2 in a context including UCAVs and lethal autonomous weapon systems.

What the report supports

Its provenance, terminology and the context described by the Panel.

What it does not establish

A specific autonomous mode, human intervention in a particular event, or harm attribution.

Source: UN Security Council, S/2021/229, 8 Mar 2021, para. 63. The cover is the primary report; it is not a forensic finding of autonomous killing.

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Capability, use, and harm are separate claims



Inference discipline

Each rung needs evidence of a different kind.
Evidence at a lower rung cannot automatically establish the next one.
The Libya report provides provenance and context; it does not close the event - mode or harm - attribution gap.

A claim ladder separates what a source says from what a conclusion would require.

Sources: UN S/2021/229, 8 Mar 2021 ; ICRC Case Study, Libya . Ladder is a conceptual evidence framework.

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Decision support can still reshape responsibility

Information and attention
System outputs influence what is noticed, prioritized or questioned.

→

Human judgment
Context, uncertainty and reasons require interpretation.

→

Ethical authority
Authorized people and institutions remain answerable.

Records connect information, reasons, authority and later contestation or review.

Human presence is not, by itself, evidence of meaningful authority or control.

Sources: ICRC, 2021; U.S. DoD Directive 3000.09, 2023. Conceptual ethics and assurance model.

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Lavender (April 2024): claims and denials

REPORTED ALLEGATIONS	IDF DENIAL / ASSERTIONS	PUBLIC AUDIT GAP
+972 / Local Call · 3 Apr 2024 Anonymous-source reporting alleged limited substantive human oversight in AI-assisted decisions.	**IDF response** · 3 Apr 2024 Denied using AI to identify operatives or predict whether a person was a militant.	**Public record** No externally inspectable model, data, version history, validation or case-level review records are available.
Status: attributed reporting, not an established system fact.	Described intelligence cross-referencing and asserted analyst examination and individual assessment. **Status:** official assertion, not independent audit.	Reuters reported on 4 Apr that the U.S. had not verified the report. **Status:** the gap proves neither side true or false.

Anonymous testimony, official safeguards and independently inspectable evidence are different kinds of claim.

Sources: +972 / Local Call, 3 Apr 2024; IDF response, 3 Apr 2024; Reuters, 4 Apr 2024.

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Asserted safeguards are not an auditable record

ASSERTED SAFEGUARD OR CLAIM	PROPOSED RECORD THAT COULD BE EXAMINED
Human examination occurred	Timestamped review record, authority and basis for the decision
The system was tested	Versioned evaluation record, scope, limits and unresolved findings
Data were responsibly governed	Data provenance, access history and change documentation
The organization is accountable	Named owners, incident log, audit access and corrective action record

A public evidence gap is a limit on verification; it is not proof that an allegation is true or false.

Source context: Reuters, 4 Apr 2024 . Record categories are proposed assurance evidence, not a claim about access to a particular system.

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Meaningful control is a set of capabilities

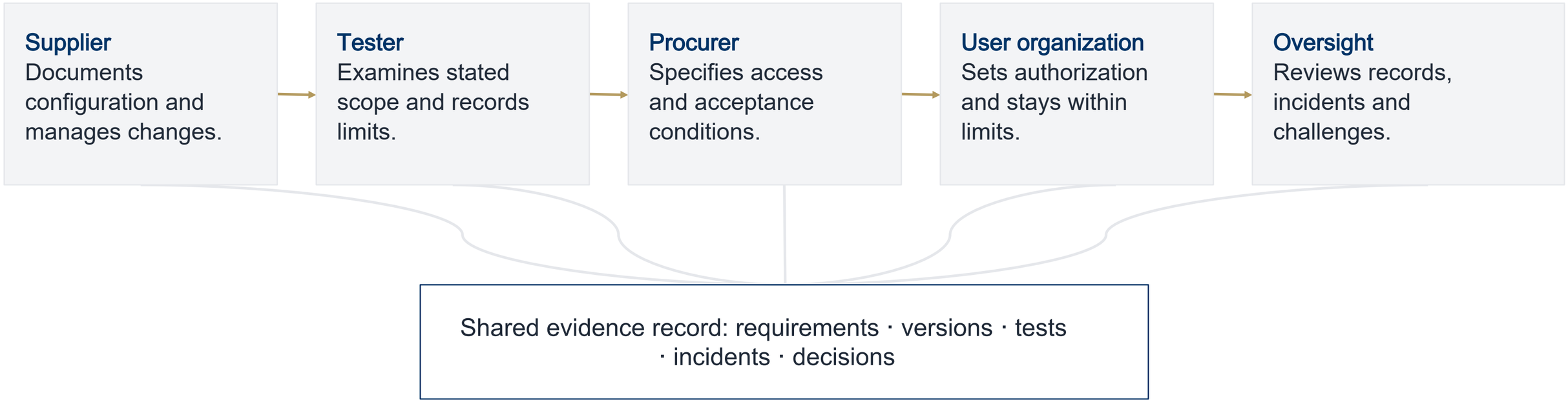
CAPABILITY	WHAT MUST BE TRUE	EVIDENCE TRACE
Understanding	System status, limits and uncertainty are intelligible.	Interface and training record
Scope	Context and boundaries are explicit.	Authorization and configuration record
Adequate time	Authority can assess rather than merely endorse.	Time-allocation assessment
Override authority	An authorized person can intervene or deactivate.	Role and access-control record
Practical intervention	The intervention path works in the stated setting.	Test and incident record
Institutional review	Material changes can trigger renewed review.	Review and version history

Meaningful control is not a single switch, a job title, or mere human presence.

Sources: ICRC, 2021; U.S. DoD Directive 3000.09, 2023. Capability table is a conceptual assurance tool.

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Responsibility crosses organizational boundaries

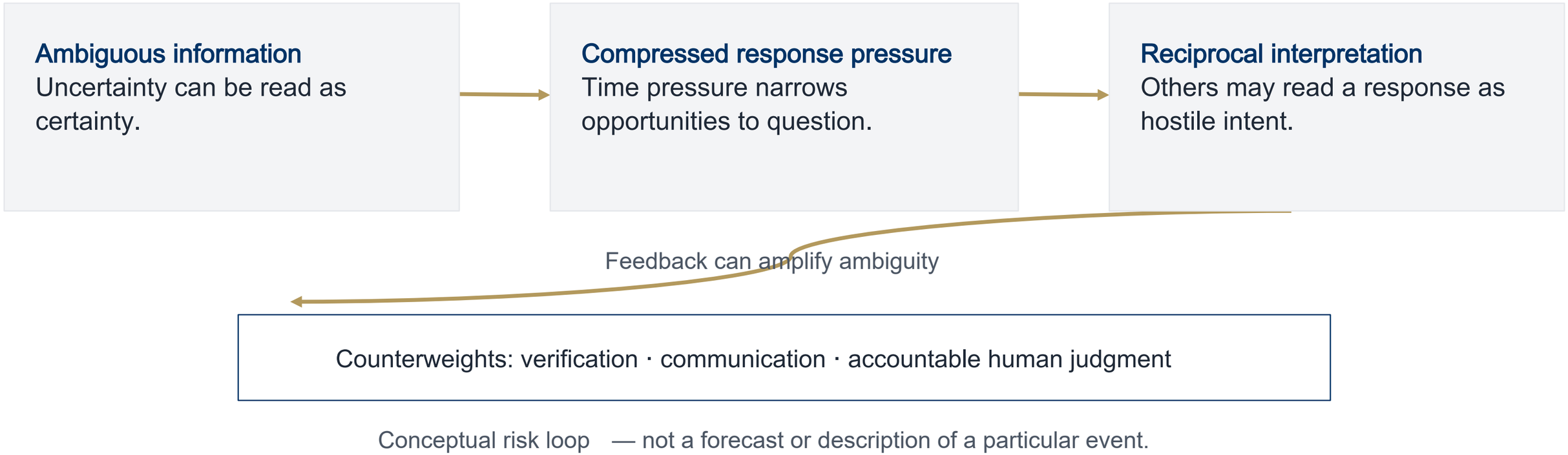


Role mapping clarifies evidence ownership; it does not allocate definitive legal liability.

Sources: U.S. DoD Directive 3000.09, 2023; NATO AI Strategy, 2024. Conceptual responsibility map.

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Compressed decisions can increase escalation risk

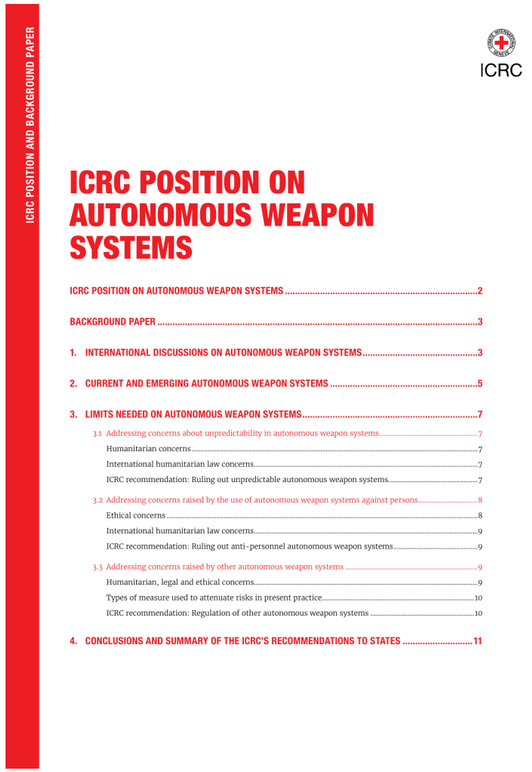


Conceptual risk loop — not a forecast or description of a particular event.

Source: ICRC, Position on Autonomous Weapon Systems, 12 May 2021. Diagram synthesizes concerns about predictability, human control and escalation.

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The ICRC proposes prohibitions and restrictions


ICRC POSITION AND BACKGROUND PAPER

ICRC

ICRC POSITION ON AUTONOMOUS WEAPON SYSTEMS

ICRC POSITION ON AUTONOMOUS WEAPON SYSTEMS 2
BACKGROUND PAPER 3
1. INTERNATIONAL DISCUSSIONS ON AUTONOMOUS WEAPON SYSTEMS 3
2. CURRENT AND EMERGING AUTONOMOUS WEAPON SYSTEMS 5
3. LIMITS NEEDED ON AUTONOMOUS WEAPON SYSTEMS 7
3.1 Addressing concerns about unpredictability in autonomous weapon systems 7
Humanitarian concerns 7
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ICRC recommendation: Ruling out unpredictable autonomous weapon systems 7
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3.3 Addressing concerns raised by other autonomous weapon systems 9
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Types of measure used to attenuate risks in present practice 10
ICRC recommendation: Regulation of other autonomous weapon systems 10
4. CONCLUSIONS AND SUMMARY OF THE ICRC'S RECOMMENDATIONS TO STATES 11

ICRC position paper · May 2021

Proposed new legally binding rules — not a universal existing ban

Prohibit

Unpredictable autonomous weapon systems

Systems designed or used to apply force against persons

Restrict other systems

Limits on scope, duration, scale and situation

Effective human supervision, intervention and deactivation

This is an ICRC policy proposal; states may take other positions.

Source: ICRC, Position on Autonomous Weapon Systems, 12 May 2021. Cover reproduced as the original public position paper.

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Human control is a humanitarian concern

Historical ICRC perspective · Full source video

Human judgment
Reasons and context remain visible.

→

Meaningful control
Authority can intervene or stop.

→

Accountability
Records permit review and redress.

Historical humanitarian argument, not evidence of current deployment or performance.

Source video: ICRC-linked, “The rise of autonomous weapons: robots fighting wars” · Historical perspective.

DoDD 3000.09 defines a U.S. policy framework

The directive frames institutional responsibilities; it is not international law.

Policy element	Institutional expectation	Limit on the claim
Human judgment	Appropriate levels of human judgment over the use of force.	A policy requirement; not a claim that every AI use has the same review path.
Assurance and interface	Rigorous hardware and software V&V, realistic test and evaluation, and clear status and role information.	Evidence must match the system and its intended context.
Certain AWS review	Senior review before formal development and again before fielding for certain autonomous weapon systems.	Specified exceptions exist; this is not senior review of all AI.

Source: U.S. DoD Directive 3000.09, effective 25 January 2023. Changes in scope or algorithms can trigger renewed review.

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Legal review and engineering assurance are complementary

Different questions need different evidence, owners, and records.

Lens	Core question	Illustrative evidence and boundary
Weapons legal review	For parties to Additional Protocol I, Article 36 requires review of a new weapon, means, or method of warfare.	Legal analysis and records of the system under review. It does not make every future use lawful.
Engineering assurance	What can the system do, where can it fail, and how are limits evidenced?	Documented V&V, test evidence, version history, and operational constraints.
Authorized use and governance	Who has authority, information, time, and practical ability to intervene?	Role design, training, access, escalation, and audit records; human presence alone is insufficient.

Source: Additional Protocol I, Article 36 (ICRC treaty database). Article 36 is an obligation for parties to that Protocol, not a universal rule for every state.

NATO principles need operational evidence

NATO’s 2024 strategy lists six responsible-use principles. The evidence and owners below are proposed mappings, not implementation claims.

2024 principle	Proposed evidence artifact	Accountable function
Lawfulness	Documented legal and policy basis for the intended context.	Authorized legal and policy review.
Responsibility and accountability	Named decision rights and an escalation record.	Program authority.
Explainability and traceability	Version, data, and decision-record provenance.	System assurance lead.
Reliability	Test evidence, limits, and monitored outcomes.	Independent test function.
Governability	Pause, rollback, and incident-response authority.	Operational owner.
Bias mitigation	Data-quality review and documented challenge process.	Data steward and reviewer.

Source: NATO, Summary of the Revised AI Strategy, 10 July 2024. Institutional strategy, not a binding treaty.

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Procurement should buy auditability, not just capability

Proposed acquisition terms translate assurance needs into access and record-keeping.

Proposed term	Why it matters	Evidence to request
Data and model rights	Reviewers need enough provenance and access to assess claimed limits.	Data lineage and model documentation.
Independent test access	Supplier statements alone do not establish assurance.	Test protocol, results, and limitations.
Version and change control	Claims can change when software, data, or context changes.	Release log and re-review trigger.
Monitoring, incidents, and exit	Accountability continues after award and must survive withdrawal.	Incident cooperation, export, and transition plan.

Learning artifact, not contract legal advice. Informed by DoDD 3000.09 (2023) and NATO's 2024 AI strategy.

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Secrecy and accountability create a governance tension

Sensitive information can be protected while authorized review remains possible.

Restricted operational records	Protect sensitive source material and personal information through need-to-know access. Restriction should be specific, time-bounded, and logged.
Authorized independent review	Permit qualified reviewers to inspect versions, evidence, decisions, and incidents under appropriate safeguards.
Aggregate public accountability	Disclose governance methods, review structures, and high-level findings without exposing protected detail.

Proposed access architecture. NATO (2024) emphasizes explainability, traceability, responsibility, and accountability; it does not prescribe this diagram.

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A fictional procurement makes the trade-offs concrete

Fictional teaching case — not a real program, performance claim, or operational system.

Fictional customer: HarborWorks

Seeks analytics to prioritize inspection of noncombat maintenance work orders.
Provides historical work orders, inspection notes, and parts records.
Human maintenance authority retains work authorization and safety release.

Fictional supplier offer

Offers a prioritization service and requests limited data transfer.
Customer requires version and change records, validation access, incident cooperation, and a suspension path.

The procurement choice is not “buy or do not buy.” It is whether evidence, authority, and rollback rights travel with the service.

Case framing uses the GAO 2022 maintenance evaluation gap as an evidence caution, not as a claim about this fictional service.

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Deployment gates connect evidence to decision rights

Proposed gates for the fictional HarborWorks maintenance service. Each gate makes a reversible decision explicit.

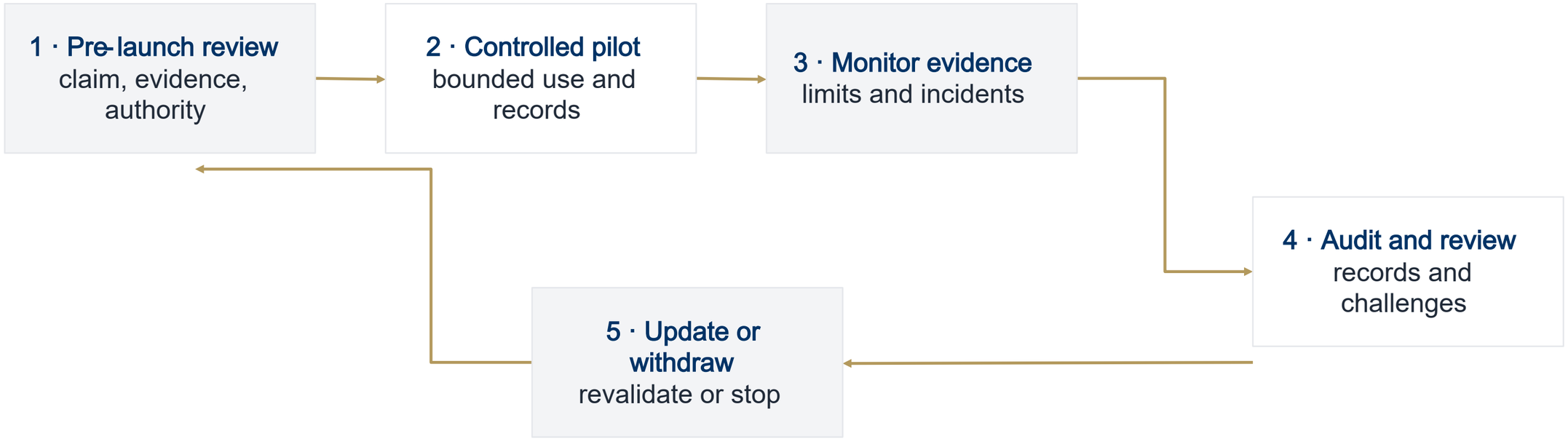
Gate	Decision owner	Evidence required to enter	Rollback or hold right
Data readiness	Data steward	Provenance, quality limits, and permitted-use record.	Quarantine data or decline transfer.
Technical validation	Independent evaluator	Documented results, limitations, and version identified.	Halt pilot pending evidence.
Workflow fit	Maintenance authority	Inspection, escalation, and human work-authorization process.	Return to the standard process.
Audit readiness	Procurement and oversight	Access, change, incident, and record-retention arrangements.	Suspend service or access.

Proposed teaching matrix for a noncombat procurement scenario. It creates no operational decision rule or performance thresho ld.

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Responsible deployment is a reversible lifecycle

A responsible lifecycle preserves authority to pause, change, or withdraw when evidence changes.



Proposed lifecycle: changes in system, data, or context should reopen the evidence and authority question.

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Responsibility cannot be delegated to a score

AI ≠ autonomy
Demo ≠ deployment
Contract ≠ public value

Allegation ≠ finding
Human presence ≠ effective control
Evidence, authority, and accountability must remain connected.

Further reading: ICRC, Position on autonomous weapon systems (May 2021)

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Case evidence and original documents

Compact, clickable primary and contemporaneous sources. Labels retain the evidence boundary used in the lecture.

Case	Readable source link	Evidence boundary
Project Maven	DoD memo (2017) / award notice (2024)	Analyst support origin; award value is not validated performance.
Maintenance analytics	GAO GAO-23-105556 (2022)	Pilot targets were not demonstrated benefits.
Patriot (2003)	Defense Science Board public summary (2005)	Historical conventional automation analogue, not modern ML.
DARPA ACE / X-62A	DARPA controlled-test announcement (2024)	Research and flight demonstration are not deployment evidence.
Libya (2021)	UN Panel report, paragraph 63	Does not establish a specific autonomous mode or harm attribution.
Lavender (2024)	+972 / Local Call investigation / IDF response	Attributed allegations and denial; no public independent technical audit.

Use the linked originals alongside their stated date, author, and scope; source existence does not settle every contested cla im.

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Law, policy, and teaching media

Source category matters: international rules, institutional frameworks, and ICRC proposals have different legal status.

Category	Readable source link	Use and boundary
International IHL	ICRC IHL overview / customary Rule 1, Rule 14, Rule 15	Rules on distinction, proportionality, and precautions; context-specific judgment.
Treaty legal review	ICRC treaty database: Additional Protocol I, Article 36	Review obligation for parties to the Protocol; not a finding about every use.
ICRC proposal	Position on autonomous weapon systems (2021)	Proposes new legally binding rules; not an existing universal ban.
U.S. institutional policy	DoDD 3000.09 (2023)	Institution-specific human-judgment and review framework, not treaty law.
NATO institutional strategy	Revised AI strategy summary (2024)	Six responsible-use principles; not a binding treaty or implementation proof.
Official source video	DARPA ACE and USAF X-62A — full source video	Test perspective; not evidence of deployment or safety certification.
Official source video	ICRC historical perspective — full source video	Historical humanitarian perspective; not empirical technology proof.

Video links are source media, not embedded clips. The excluded ICRC video is not listed.